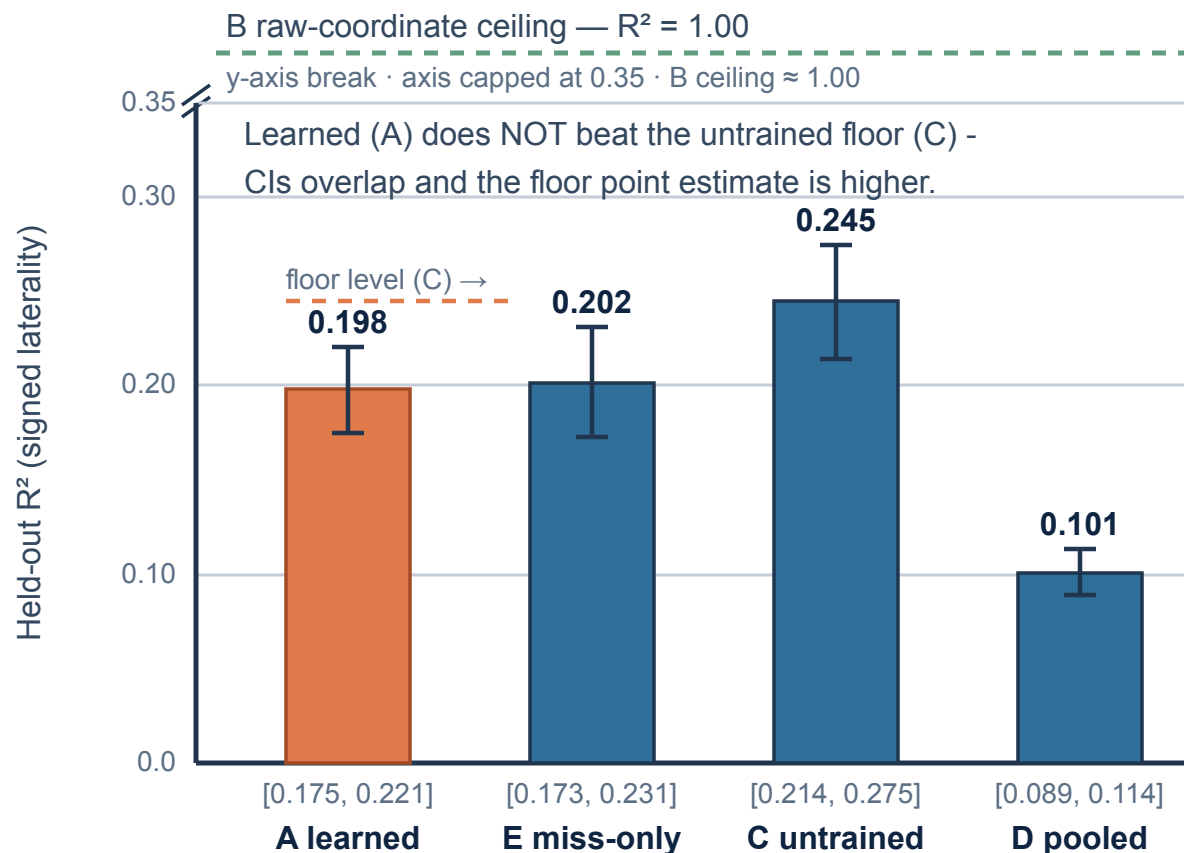


Bilateral symmetry does not emerge: an informative null

Bars = mean held-out R^2 with 95% CI whiskers · 626 modeled sequences / 93 source videos · source-disjoint CV (10 seeds) · SVD ridge

Held-out signed-laterality decodability (R^2)



Verdict

INFORMATIVE NULL

Pre-registered gates: all FAIL

A-C < 0.05 (margin gate)

A < 80% of B (ceiling gate)

sign 0.55 < 0.75 (sign gate)

Two corroborating checks

Anatomical-mirror slope = -0.70 (no sign flip).

Missingness-only (E) matches learned (A) -
apparent signal is partly a left/right
visibility artifact.

All results are transductive (internal validity only; the encoder was trained on the evaluated sequences).

Source video is the independent unit; folder labels are dataset annotations, not diagnoses.